

Software Requirements Specification

Unified Vehicle Marketplace & Real-Time Auction Engine

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Table of Contents

Document Governance

1. Document Control & Executive Summary

- 1.1 Purpose
- 1.2 System Scope
- 1.3 Business Objectives
- 1.4 Hybrid Value Proposition
- 1.5 Definitions & Acronyms

2. User Architecture & Role-Based Access Control (RBAC)

- 2.1 RBAC Model Principles
- 2.2 Persona Definitions
- 2.3 Permission Matrix (Representative Extract)

3. High-Level Functional Requirements — Module Breakdown

- Module A — Vehicle Ingestion & Catalog Management
- Module B — Advanced Search & Discovery Engine
- Module C — Core Marketplace Functionality
- Module D — Real-Time Auction & Bidding Core
- Module E — Financial, Escrow & Transaction Engine
- Module F — Post-Sale Operations & Logistics

4. UI/UX Strategy & User Journey Flows

- 4.1 Design System Mandates
- 4.2 Platform Surface Requirements
- 4.3 User Journey — Listing a Car (Private Seller)
- 4.4 User Journey — Bidding on a Car (Trade Buyer)
- 4.5 Journey-Level Experience Requirements

5. Non-Functional Requirements (NFRs)

- 5.1 Performance
- 5.2 Scalability & Availability
- 5.3 Security & Compliance
- 5.4 Observability, Auditability & Operability

6. Proposed Technical Architecture & Infrastructure Stack

- 6.1 Architecture Style
- 6.2 Frontend & Mobile
- 6.3 Backend & Real-Time Services
- 6.4 Database Layer — Polyglot Persistence
- 6.5 Infrastructure & Delivery
- 6.6 Build vs. Buy Summary

7. Requirements Traceability & Acceptance

- Assumptions & Dependencies
- Open Items for Architecture Review Board

Document Governance

Revision History

Version	Date	Author	Description
0.1.0	2026-06-20	Product Architecture Office	Initial scoping draft
0.5.0	2026-07-01	Product Architecture Office	Module decomposition, NFR baselines
1.0.0	2026-07-13	Product Architecture Office	Full draft for ARB review
1.1.0	2026-07-13	Product Architecture Office	Backend technology baseline revised to PHP 8.4 / Laravel 12 (Octane/ Swoole auction core, Reverb WebSocket tier, Horizon workers); bid log re-platformed to Redis Streams

Approvals

Role	Name	Signature	Date
Chief Product Officer	—	Pending	—
Chief Technology Officer	—	Pending	—
Head of Compliance & Risk	—	Pending	—
Head of Security (CISO)	—	Pending	—

Referenced Documents

Ref	Document
[R1]	ISO/IEC/IEEE 29148:2018 — Requirements Engineering
[R2]	ISO 3779:2009 — Road vehicles: Vehicle Identification Number (VIN) content and structure
[R3]	UK GDPR & Data Protection Act 2018
[R4]	PCI-DSS v4.0 — Payment Card Industry Data Security Standard
[R5]	FCA Handbook — CONC (Consumer Credit sourcebook) & Consumer Duty (PRIN 12)
[R6]	NAMA (National Association of Motor Auctions) Vehicle Grading Standard

Ref	Document
[R7]	WCAG 2.2 AA — Web Content Accessibility Guidelines
[R8]	Money Laundering Regulations 2017 (as amended) — KYC/KYB obligations

Conventions

- Requirement identifiers follow the pattern `<TYPE>--<MODULE>--<NNN>` (e.g., `FR-D-012`, `NFR-P-003`).
- Priorities use MoSCoW notation: **M** (Must), **S** (Should), **C** (Could), **W** (Won't — this release).
- “The Platform” refers to the complete HMP system. “The Auction Core” refers to Module D services specifically.
- The verbs **shall** (mandatory), **should** (recommended), and **may** (optional) carry their RFC 2119 meanings.

1. Document Control & Executive Summary

1.1 Purpose

This SRS defines the complete functional, non-functional, and architectural requirements for the **Hybrid Motors Platform (HMP)** — a unified digital automotive commerce system combining:

1. A **consumer-facing vehicle marketplace** (reference architecture: AutoTrader UK) — advertising-led discovery, dealer storefronts, valuations, and financing.
2. A **trade-grade, real-time remarketing auction engine** (reference architecture: BCA Group) — multi-lane simulcast auctions, proxy bidding, inspection-backed vehicle grading, and complex fee settlement.

The document is the contractual baseline for engineering delivery, vendor selection, security review, and compliance sign-off.

1.2 System Scope

In scope:

- Vehicle ingestion, enrichment (VIN/VRM decode, provenance checks), and catalog management for both retail listings and auction lots.
- Consumer marketplace: search & discovery, dealer storefronts, private listings, valuations, finance integration, lead management, and advertising/SEO monetization.
- Auction operations: physical-simulcast and digital-only sales, multi-lane real-time bidding, proxy bidding, reserve management, auctioneer console, and lot lifecycle management.
- Financial engine: deposits, escrow-style secured funds handling, buyer/seller fee computation, invoicing, VAT (including margin scheme) handling, trade credit facilities, and multi-party settlement.

- Post-sale operations: logistics/transport booking, vehicle release, title/V5C transfer tracking, and dispute/assurance workflows.
- Web (responsive), iOS, and Android clients; internal operational consoles; public and partner APIs.

Out of scope (this release):

- Physical auction-hall hardware (rostrum AV, lane signage controllers) beyond defined integration APIs.
- In-house underwriting of consumer finance (the Platform brokers/integrates; it does not lend).
- New-vehicle OEM configurator commerce.
- Non-UK regulatory localization (architecture shall not preclude it — see NFR-S-014).

1.3 Business Objectives

ID	Objective	KPI / Target
BO-01	Capture full vehicle lifecycle value: one asset, many monetization events (retail listing → part-exchange → trade auction → retail again)	≥ 25% of auctioned vehicles re-listed on marketplace within 30 days
BO-02	Diversified revenue: dealer subscriptions, listing/ad products, buyer & seller auction fees, finance commission, logistics margin, data products	No single revenue stream > 40% of total by Year 2
BO-03	Trade liquidity: reduce average days-to-sell for trade stock	Median lot cycle (ingest → settled) < 9 days
BO-04	Consumer trust: inspection-graded, provenance-checked inventory as a differentiator vs. classified-only competitors	≥ 60% of listings carrying a verified condition grade
BO-05	Operational leverage: digital-first auctions reducing per-lot operating cost	≥ 70% of bids placed via digital channels within 12 months

1.4 Hybrid Value Proposition

The strategic thesis of HMP is that the marketplace and the auction engine are **not two products but one liquidity flywheel**:

- **Supply-side:** A dealer's part-exchange that fails retail criteria is routed frictionlessly into the next auction sale (single ingestion, dual channel). A private seller can choose “instant offer → auction disposal” instead of a classified listing.
- **Demand-side:** Trade buyers replenish retail forecourt stock from auctions and immediately re-list on the marketplace with inspection data, imagery, and valuation carried forward — zero re-keying.

- **Data moat:** Every transaction (retail asking price, achieved auction hammer price, condition grade, days-to-sell) feeds a proprietary valuation model whose accuracy compounds with volume, powering the consumer valuation tool, dealer stocking intelligence, and reserve-price recommendations.

1.5 Definitions & Acronyms

Term	Definition
VRM	Vehicle Registration Mark (UK number plate)
VIN	Vehicle Identification Number (ISO 3779)
Lot	A vehicle entered into a specific auction sale event
Hammer Price	Final accepted bid price, exclusive of fees
Reserve	Confidential minimum price authorized by the seller
Proxy Bid	A buyer's confidential maximum; the system bids incrementally on their behalf
Simulcast	A physical-lane auction broadcast live with remote digital bidders participating in real time
Provisional Sale	Highest bid below reserve; referred to seller for accept/decline/counter
CAP/Glass's-class Data	Third-party trade valuation reference data
V5C	UK vehicle registration certificate (logbook)
KYB/KYC	Know Your Business / Know Your Customer (AML obligations)
SLA/SLO/SLI	Service Level Agreement / Objective / Indicator
DDD	Domain-Driven Design

2. User Architecture & Role-Based Access Control (RBAC)

2.1 RBAC Model Principles

- **REQ RBAC-001 (M):** The Platform shall implement RBAC per **NIST RP-BAC** semantics with hierarchical roles, augmented by **ABAC (attribute-based) policy conditions** (e.g., “Trade Buyer MAY bid IF account_status = active AND credit_available ≥ lot_deposit_requirement AND KYB_verified = true”).
- **REQ RBAC-002 (M):** All authorization decisions shall be externalized to a central Policy Decision Point (PDP) — see §6 (proposed: OPA/Cedar) — never hardcoded in service logic.

- **REQ RBAC-003 (M):** Roles shall be scoped to an **Organization** entity (dealership group → rooftop → user), supporting multi-rooftop franchise structures with delegated administration.
- **REQ RBAC-004 (M):** Every privileged action shall emit an immutable audit event (actor, role, resource, decision, policy version, timestamp) retained ≥ 7 years.
- **REQ RBAC-005 (S):** The Platform shall support Just-In-Time elevation with mandatory reason capture and time-boxed grants for Super Admin operations (break-glass model).

2.2 Persona Definitions

P1 — Private Buyer

Consumer researching and purchasing a vehicle. Anonymous browsing permitted; account required for saved searches, alerts, messaging, finance applications, and direct-buy checkout.

- **Capabilities:** search/browse, save searches & vehicles, compare, message sellers, request finance quotes (soft-search only), reserve/purchase via direct-buy, book test drives, leave verified purchase reviews.
- **Explicitly denied:** trade auction bidding, wholesale price visibility, condition-report internals beyond consumer summary.

P2 — Private Seller

Consumer listing a personally owned vehicle or requesting instant-offer disposal.

- **Capabilities:** create/manage one active listing (configurable cap), VRM-lookup-assisted listing creation, receive/respond to messages and offers, accept instant-offer routing into trade auction (with consent workflow), track sale status and payout.
- **Constraints:** identity verification (KYC-lite) required before listing goes live; payout only to a verified bank account in the seller's name.

P3 — Franchise Dealership (Organization role bundle)

A retail dealer organization with sub-roles: **Org Admin**, **Sales Manager**, **Sales Executive**, **Stock Controller**, **Finance & Insurance (F&I) Manager**.

- **Capabilities:** dealer storefront management, bulk stock ingestion (feed/DMS integration), subscription & ad-product management, lead inbox/CRM integration, valuation & market intelligence tools, part-exchange appraisal, consignment of stock into auctions (dual-channel routing), auction buying under trade terms.
- **Delegated admin:** Org Admin manages users, rooftops, permissions, and spending limits within the organization.

P4 — Independent Trade Buyer

Sole trader or small motor-trade business buying at auction to retail independently.

- **Capabilities:** auction catalog access (including full condition reports and grade detail), live bidding, proxy bidding, "Buy Now" on fixed-price trade stock, trade credit application, transport booking, assurance/dispute claims.

- **Gating:** account activation requires KYB verification (Companies House / sole-trader evidence), motor-trade insurance evidence, deposit or credit line establishment. Bidding entitlement is dynamically constrained by available funds/credit (ABAC condition).

P5 — Platform Auctioneer (Internal)

Licensed auction operator conducting live sales.

- **Capabilities:** rostrum console — lane control (start/pause/skip lot), bid acknowledgment for hall bids, reserve visibility for lots in their lane, provisional-sale referral, hammer/withdraw actions, bid retraction handling with mandatory reason codes.
- **Constraints:** cannot alter fees, cannot see buyer identities beyond paddle/bidder number during the sale (anti-collusion), all rostrum actions are video/audit synchronized.

P6 — Vehicle Inspector (Internal / Contracted)

Field or site-based technician producing condition reports.

- **Capabilities:** mobile-first inspection app — guided capture workflow (photos, videos, damage annotation, tyre depths, service history verification, diagnostics/OBD readout attach), grade calculation review, report submission and amendment (pre-publication only).
- **Constraints:** write access limited to assigned inspection tasks; post-publication amendments require Quality Supervisor countersign and produce a new report version (never in-place mutation).

P7 — Global Super Admin (Internal)

Platform operations root authority.

- **Capabilities:** tenant/org management, role & policy administration, fee-schedule publication, feature-flag control, sanctions/fraud actioning (account freeze, listing takedown), financial reconciliation overrides.
- **Constraints:** dual-control (four-eyes) enforcement for destructive/financial overrides; all sessions hardware-key MFA; every action justified via linked ticket reference (REQ RBAC-005).

Supporting internal roles (summary)

Role	Purpose
Customer Support Agent	Impersonation-view (read-only, consented, watermarked), case management
Trust & Safety Analyst	Fraud queue triage, listing moderation, ban management
Finance Operations	Settlement runs, refunds, VAT adjustments, ledger reconciliation
Logistics Coordinator	Transport job assignment, exception handling
Data & Pricing Analyst	Valuation model monitoring, reserve recommendation oversight

2.3 Permission Matrix (Representative Extract)

Capability	P1	P2	P3	P4	P5	P6	P7
Browse retail listings	✓	✓	✓	✓	✓	—	✓
View wholesale/ auction catalog	—	—	✓	✓	✓	—	✓
View full condition report internals	—	Own vehicle	✓	✓	✓	Assigned	✓
Place live/ proxy bid	—	—	✓ *	✓ *	—	—	—
See reserve price	—	Own lot	Own lots	—	Lane lots	—	✓
Publish fee schedule	—	—	—	—	—	—	✓ (4-eyes)
Hammer / withdraw lot	—	—	—	—	✓	—	Break-glass
Amend published inspection	—	—	—	—	—	Countersigned	✓ (4-eyes)

* Subject to ABAC funding/verification conditions (§2.2 P4).

3. High-Level Functional Requirements — Module Breakdown

Module A — Vehicle Ingestion & Catalog Management

A.1 Vehicle Identity & Enrichment

- **FR-A-001 (M):** The Platform shall create a single canonical **Vehicle Master Record (VMR)** per physical vehicle, keyed by VIN, with VRM as a mutable alias. All listings and lots reference the VMR; the same physical vehicle shall never fork into divergent catalog records.

- **FR-A-002 (M):** VRM lookup shall integrate the **DVLA Vehicle Enquiry Service (VES)** API to resolve make, model, colour, fuel type, engine capacity, CO₂, Euro status, MOT expiry, and tax status; response cached with a 24-hour TTL and provenance timestamp.
- **FR-A-003 (M):** VIN decoding shall validate structure per ISO 3779 (WMI/VDS/VIS, check digit where applicable) and enrich against a commercial vehicle-data provider (CAP HPI / Autovista class) for derivative-level taxonomy: trim, transmission, drivetrain, factory options, battery capacity & usable kWh for BEV/PHEV.
- **FR-A-004 (M):** Provenance screening shall execute on ingestion and re-execute pre-publication and pre-sale: outstanding finance markers, stolen (PNC-sourced feeds via provider), insurance write-off category (A/B/S/N), mileage anomaly (National Mileage Register class), plate transfer history, and import status. A failed check shall hard-block publication with a triage case raised.
- **FR-A-005 (M):** MOT history shall be ingested via the DVSA MOT History API, with advisories surfaced structurally (not free text only) and mileage points fed to the odometer-anomaly detector.
- **FR-A-006 (S):** The taxonomy service shall maintain a versioned make/model/derivative tree with effective-dating, so historical listings remain queryable against the taxonomy version at listing time.

A.2 Media Handling

- **FR-A-010 (M):** The media pipeline shall accept up to 100 stills, 5 videos (≤ 4 min each), and one 360° spin set (interior panorama + exterior turntable of ≥ 36 frames) per vehicle, with client-side resumable upload (tus/S3 multipart) tolerating mobile network interruption.
- **FR-A-011 (M):** All imagery shall be transcoded server-side into a responsive rendition ladder (AVIF/WebP with JPEG fallback; DPR 1x–3x), served via CDN with signed URLs for pre-publication assets.
- **FR-A-012 (M):** Automated media QA shall run on ingest: blur/exposure scoring, VRM plate detection with optional auto-anonymization overlay, duplicate-frame detection, and prohibited-content screening. Failures return structured remediation guidance to the uploader.
- **FR-A-013 (S):** 360° exterior capture shall support **damage hotspot annotation**: inspectors pin damage markers to frame coordinates; the viewer renders interactive hotspots linking to the corresponding condition-report line and close-up photo.
- **FR-A-014 (S):** The Platform shall support guided capture in the mobile apps (overlay silhouettes per shot type: front-3/4, dashboard, odometer, tyre, etc.) with per-shot auto-validation, referencing the shot list to the listing-quality score (FR-C-014).
- **FR-A-015 (C):** AI background replacement/standardization for dealer imagery, applied only with explicit dealer opt-in and an on-image “enhanced” disclosure flag.

A.3 Condition Grading Logic

- **FR-A-020 (M):** The Platform shall implement a **NAMA-aligned grading engine** producing: (a) a **Condition Grade 1–5** (1 = best) computed deterministically from itemized damage entries via a published costing matrix (damage type $\times$ panel $\times$ severity $\rightarrow$ repair-cost band), and (b) a separate **Mechanical Grade** (drive/run status, warning lamps, service evidence).
- **FR-A-021 (M):** The grading matrix shall be a versioned configuration artifact; every published grade stores the matrix version used, enabling exact recomputation and audit. Matrix changes require dual approval and take effect only for inspections started after the effective timestamp.

- **FR-A-022 (M):** The inspection workflow (Inspector persona) shall enforce a mandatory checklist by vehicle category (car / LCV / motorcycle / BEV) — BEV inspections shall additionally capture State-of-Health (SoH) readout, charge-cable inventory, and charging-session verification.
- **FR-A-023 (M):** Each condition report shall be an immutable, versioned document with digital signature of the inspecting user, geolocation & timestamp of capture, and a consumer-safe summary projection (P1-visible) distinct from the trade detail projection (P3/P4-visible).
- **FR-A-024 (S):** An **assurance product** flag (“HMP Assured”, modeled on BCA Assured) shall be attachable to lots meeting inspection-freshness (< 14 days) and mechanical criteria, driving the post-sale claims workflow (FR-F-020).
- **FR-A-025 (S):** Computer-vision damage pre-detection shall propose damage entries from imagery for inspector confirmation (human-in-the-loop mandatory; CV output alone shall never publish a grade).

A.4 Dual-Channel Catalog & Stock Routing

- **FR-A-030 (M):** A VMR shall be routable to one active sales channel at a time — Retail Listing, Fixed-Price Trade (“Buy Now”), or Auction Lot — with a channel-switch workflow that atomically closes the outgoing channel (withdrawing the listing / removing the lot) before opening the incoming one.
- **FR-A-031 (M):** Dealer bulk ingestion shall support: SFTP/CSV feed, DMS integrations (Keyloop/CDK, Pinewood class) via adapter framework, and a public Stock API (REST + webhook callbacks), with per-feed field-mapping templates and dry-run validation reports.
- **FR-A-032 (S):** Rule-based auto-routing: dealers may define policies such as “part-exchange, age > 8 years OR grade $\geq 4 \rightarrow$ consign to next weekly auction, reserve = CAP Clean – 5%”.

Module B — Advanced Search & Discovery Engine

- **FR-B-001 (M):** Full-text and faceted search over the retail catalog and (for trade personas) the auction catalog, with facets including: make, model, derivative, price / monthly-payment band, year, mileage, fuel, transmission, body type, colour, doors, seats, CO₂/tax band, BEV range & battery health band, distance-from-postcode, condition grade, sale channel, auction date, and seller type. Facet counts shall reflect the currently applied filter set (post-filter aggregation).
- **FR-B-002 (M):** Search latency SLO: $P95 \leq 300$ ms server-side for the standard query envelope (§5). Index freshness: catalog mutations visible in search ≤ 5 s (retail), ≤ 1 s for auction-lot state changes (e.g., “sold” badge).
- **FR-B-003 (M):** Geospatial search by postcode/radius with drive-time banding (isochrone approximation acceptable), and national fallback with delivery-eligible flagging.
- **FR-B-004 (M): Monthly-payment search:** users may search by budget/month; the engine shall compute indicative PCP/HP monthly figures per vehicle (rate cards per lender panel, configurable deposit/term/mileage) and filter/sort on them. Representative-example and APR disclosures per FCA CONC 3 shall accompany every payment figure (see FR-E-030).
- **FR-B-005 (M):** Saved searches with multi-channel alerting (push, email, in-app) on new/changed matches; alert fan-out ≤ 5 minutes from listing publication; per-user frequency capping and one-click unsubscribe (PECR compliant).
- **FR-B-006 (M):** Synonym, typo-tolerance, and automotive-domain query understanding: “estate” $\leftrightarrow$ “touring” $\leftrightarrow$ “avant”-class synonym sets per manufacturer, VRM direct lookup from the search bar, and structured query parsing (“blue automatic golf under 15k near LS1”).

- **FR-B-007 (S): Recommendations:** (a) similar-vehicle recommendations on the detail page (vector similarity over derivative, price, spec, and imagery embeddings); (b) personalized homepage rails from consented behavioral signals; (c) trade-buyer lot recommendations from purchase history and stocking profile (“dealers like you retail this derivative in 22 days at +£1,400 margin”).
- **FR-B-008 (S):** Ranking shall combine lexical relevance, listing-quality score, freshness, price competitiveness vs. valuation, and (for sponsored slots) ad auction rank — with sponsored placements visibly labeled and capped per result page (CAP/ASA compliance).
- **FR-B-009 (S):** Price-drop and market-context badges (“£800 below market average”) computed against the valuation service, recalculated on price change events.
- **FR-B-010 (C):** Natural-language conversational search assistant grounded exclusively in live catalog data, with guardrails preventing invented inventory or unverifiable claims.

Module C — Core Marketplace Functionality

C.1 Listings & Storefronts

- **FR-C-001 (M):** Listing lifecycle state machine:
Draft → PendingChecks → Live → (UnderOffer|Reserved) → Sold|Withdrawn|Expired, with every transition evented (for search, analytics, billing) and permissible transitions role-gated.
- **FR-C-002 (M):** Dealer storefront pages: branded profile, full stock grid with dealer-scoped search, opening hours, verified reviews aggregate, FCA/regulatory disclosures block, click-to-call with tracked numbers, and lead forms. Storefronts shall be SEO-indexable with structured data (schema.org `Vehicle`, `AutoDealer`, `Offer`).
- **FR-C-003 (M):** Private listings: guided creation flow from VRM lookup (pre-filled spec), photo guidance, valuation-anchored pricing suggestion, ID verification gate before go-live, and configurable listing durations with renewal.
- **FR-C-004 (M):** SEO infrastructure: server-rendered listing/search pages, canonicalization rules for faceted URLs (index whitelisted facet combinations only), XML sitemaps regenerated on publication events, Core Web Vitals budgets (§5 NFR-P-010), and 301-preservation on listing expiry to relevant search pages.
- **FR-C-014 (S): Listing-quality score** (0–100) computed from media completeness, spec completeness, description quality, provenance-check freshness — surfaced to sellers with remediation prompts and used as a ranking signal (FR-B-008).

C.2 Monetization — Subscriptions & Advertising

- **FR-C-020 (M):** Dealer subscription packages (e.g., **Core / Advanced / Premium**) as configurable bundles of: stock-slot count, media entitlements (video/360°), ranking boosts, market-intelligence access, and lead-management features. Package definitions shall be data-driven (no code deploy to change entitlements), with pro-rated upgrades/downgrades and usage metering.
- **FR-C-021 (M):** Ad products: featured placement, homepage rails, model-page sponsorship, and retargeting audiences — with impression/click/lead attribution reporting per dealer, and frequency capping. All personal-data-based targeting is consent-gated (§5 Security & Compliance).
- **FR-C-022 (S):** Self-serve dealer performance analytics: search impressions, detail-page views, lead conversion, price-position vs. market, days-to-sell benchmarks.

C.3 Transactions & Communication

- **FR-C-030 (M): Direct-buy workflow** for eligible retail stock: refundable holding deposit (fixed, e.g., £199) via PSP, distance-selling documentation pack generation, 14-day cooling-off handling per Consumer Contracts Regulations for distance sales, and handover/delivery scheduling handoff to Module F.
- **FR-C-031 (M):** In-platform messaging between buyers and sellers: threaded per-listing conversations, template quick-replies, attachment support (docs/images with AV scanning), automated PII/off-platform-payment-luring detection with Trust & Safety escalation, and full retention for dispute evidence.
- **FR-C-032 (M):** Offer management: structured offer/counter-offer objects (not free text), part-exchange declaration with VRM lookup and instant appraisal range, and dealer-side lead routing rules (round-robin / rooftop / named salesperson).
- **FR-C-033 (S):** Test-drive and appointment booking with dealer calendar integration (ICS/Graph/Google), reminders, and no-show tracking.
- **FR-C-034 (M): Valuation tool (consumer):** free VRM + mileage valuation returning private-sale, part-exchange, and instant-offer figures with confidence band; powered by the internal valuation model (§1.4) blended with licensed trade data; every valuation stored for model feedback against eventual achieved prices.

Module D — Real-Time Auction & Bidding Core

The Auction Core is the highest-criticality subsystem. It is architected as an isolated, event-sourced domain (see §6) so that marketplace traffic can never degrade live bidding.

D.1 Sale & Lot Management

- **FR-D-001 (M):** Sale-event model: a **Sale** contains 1..n **Lanes**; each Lane contains an ordered **Run List** of Lots. Sale types: `PhysicalSimulcast` (hall + online), `DigitalLive` (auctioneer-led, online-only), and `TimedOnline` (eBay-style closing timers, no auctioneer). All three shall share the same bid-processing core.
- **FR-D-002 (M):** Lot lifecycle: `Cataloged` → `Published` → `InLane` → `Open` → `(Hammered|Provisional|NotSold|Withdrawn)` → `Settled`, event-sourced with complete state history. Catalog publication shall freeze the condition-report version and fee schedule version applicable to the lot.
- **FR-D-003 (M):** Run-list management: drag-and-drop reordering (pre-sale), late-entry insertion, lot withdrawal with mandatory reason codes, and automated run-order optimization suggestions (e.g., interleave high-interest lots to sustain lane attendance).
- **FR-D-004 (M):** Reserve management: reserves are write-once per lot per sale (amendable only by the seller or authorized account manager before the lot enters `InLane`), stored encrypted, visible strictly per the RBAC matrix (§2.3), and never transmitted to any client except the auctioneer console for lots in that auctioneer's active lane.

D.2 Bid Processing Engine

- **FR-D-010 (M): Single-writer sequencing:** each live lot shall be owned by exactly one bid-sequencer instance which serializes all bid attempts (digital, proxy-generated, and auctioneer-entered hall bids) into a totally ordered, append-only bid log. Order of acceptance is the order in the log — no ambiguity, ever.
- **FR-D-011 (M):** Bid validation pipeline (all checks synchronous, pre-acceptance): bidder entitlement (ABAC: verified, funded, not suspended, not the lot's seller or an associate), increment validity vs. the lane's increment table, monotonic price check, per-bidder spend/exposure limit, and duplicate/replay rejection via client-supplied idempotency keys.
- **FR-D-012 (M): Latency budget:** intra-region bid acceptance (gateway ingress → sequencer decision → confirmation event emitted) shall complete in **< 100 ms at P99** under peak load (§5 NFR-P-001). End-to-end bidder-perceived confirmation < 250 ms P95 on a reference 4G connection.
- **FR-D-013 (M):** Increment tables shall be configurable per lane/price band (e.g., £0–1k: £50; £1k–5k: £100; £5k–20k: £250; > £20k: £500) with auctioneer ability to call a custom increment (“half-bid”) that the engine validates and records.
- **FR-D-014 (M):** Bid retraction shall be an auctioneer/admin-mediated compensating event (never deletion of log entries), with automatic state rewind to the prior valid high bid and a broadcast correction event to all clients.
- **FR-D-015 (M):** Anti-sniping (TimedOnline): a bid within the final N seconds (configurable, default 60 s) extends the lot timer by N seconds, unlimited extensions, all clients receiving the authoritative new end-time.

D.3 Automated Proxy Bidding

- **FR-D-020 (M):** Buyers may lodge a confidential maximum (proxy) per lot at any time from catalog publication until lot close. Proxies are encrypted at rest and readable by no human role (including Super Admin) — only the sequencer's proxy resolver may evaluate them.
- **FR-D-021 (M): Proxy state machine** (per lot, executed inside the sequencer to preserve total ordering):
 - **ARMED** → on competing bid: if proxy max $\geq$ required next increment, emit system bid at the *minimum amount necessary to lead* (current + increment, capped at proxy max) → **LEADING**.
 - **LEADING** → outbid: re-evaluate; if exhausted → **EXHAUSTED** (notify bidder immediately with one-tap raise action); else counter-bid → **LEADING**.
 - Two competing proxies shall resolve in a single sequenced exchange to the correct final price (higher max leads at lower max + one increment, tie broken by earliest lodgment time), emitting the intermediate bid events for the public log.
- **FR-D-022 (M):** Proxy vs. reserve interaction: a proxy shall never auto-bid above its max nor reveal the reserve; if the proxy max is below reserve, the engine bids up to the proxy max normally and the lot proceeds to **Provisional** handling as usual.
- **FR-D-023 (S):** Proxy portfolio view for trade buyers: all armed proxies, aggregate maximum exposure vs. available funds/credit, with the funding check (FR-D-011) evaluating *committed* exposure, not just the single lot.

D.4 Live Distribution (Simulcast)

- **FR-D-030 (M):** Live state distribution over **WebSockets** (WSS) with a defined protocol: server-authoritative lot state snapshots plus monotonic per-lot sequence-numbered deltas; clients detect gaps by sequence number and request snapshot resync. Protocol shall be versioned and backward compatible for one major client version.
- **FR-D-031 (M):** On reconnect (mobile network churn), a client shall restore correct lot state in ≤ 1 s via snapshot + delta replay; a bidder's in-flight bid outcome shall be determinable post-reconnect via idempotency-key lookup (no “did my bid land?” ambiguity).
- **FR-D-032 (M): Synchronized lane timers:** all countdown/decision timers are server-authoritative; clients render from (server_deadline – estimated_offset) using an NTP-style clock-offset handshake on connect and periodic re-sync, so displayed timers across all clients agree within ± 150 ms.
- **FR-D-033 (M):** Multi-lane viewing: trade clients shall subscribe to up to 6 concurrent lanes with one active bidding focus; per-lane audio/video streams delivered via low-latency protocol (WebRTC or LL-HLS, glass-to-glass ≤ 3 s for simulcast video; bid data always via the WebSocket channel and never gated on video).
- **FR-D-034 (M):** Auctioneer rostrum console: current/next lot panel, bid feed with channel origin (hall/online/proxy), reserve indicator (met/not met/distance-to-reserve band), one-tap hammer, provisional referral, withdraw, custom increment, and hall-bid entry — all actions ≤ 2 interactions and latency-instrumented.
- **FR-D-035 (M):** Degradation policy: if video fails, bidding continues (data channel independent); if a bidder's connection quality drops below threshold mid-lot, the client shall display a prominent integrity warning before accepting further bid taps.
- **FR-D-036 (S):** Spectator scale-out: non-bidding viewers may be served state via a fan-out tier (server-sent events / read-only WS) decoupled from the bidder-priority path, with bidders always on the priority tier.

D.5 Sale Conclusion

- **FR-D-040 (M):** On hammer with reserve met: atomically create the **Trade Order** (buyer, lot, hammer price, applicable fee schedule version), emit `LotSold`, freeze the bid log, and trigger Module E invoicing. On hammer below reserve: create `ProvisionalSale` with a seller decision workflow (accept / decline / counter, configurable decision SLA, default 3 business hours) and automated buyer counter-negotiation messaging.
- **FR-D-041 (M):** Full bid-log transparency artifact: for every concluded lot, an immutable, timestamped bid history (bidder-pseudonymized) shall be retained and exportable for dispute resolution and regulatory audit.
- **FR-D-042 (S):** Post-sale instant analytics to seller: achieved vs. reserve vs. valuation guidance, bidder depth (unique bidders, bids), and channel mix.

Module E — Financial, Escrow & Transaction Engine

E.1 Ledger Foundation

- **FR-E-001 (M):** All money movements shall be recorded in an internal **double-entry ledger** (immutable journal, debits = credits enforced at write time), with sub-ledgers per counterparty (buyer, seller, platform revenue, VAT control, PSP settlement, client-money segregation). The ledger is the single source of financial truth; PSP and bank statements reconcile to it (FR-E-050).

- **FR-E-002 (M):** Client funds (deposits, sale proceeds pending payout) shall be held segregated from platform operating funds, via a safeguarding/client-money account structure with the regulated PSP/EMI partner, and never commingled in the ledger or at bank level.

E.2 Deposits, Payments & Escrow-Style Settlement

- **FR-E-010 (M):** Trade buyer funding models: (a) refundable security deposit establishing a bidding limit multiplier, (b) approved **trade credit line** (FR-E-020), (c) pay-per-purchase via Open Banking (Faster Payments) or card. Available headroom = (deposit multiplier + credit available) – committed exposure (won-unpaid lots + armed proxy maxima per FR-D-023).
- **FR-E-011 (M):** Direct-buy consumer flow (FR-C-030): PSP card/Apple Pay/Google Pay with 3DS2 (SCA), holding-deposit capture, and automated refund on valid cooling-off cancellation. The Platform shall be **PCI-DSS SAQ-A scoped**: card data touches PSP-hosted fields/SDKs only; no PAN shall ever transit or rest on Platform infrastructure.
- **FR-E-012 (M):** Escrow-style flow for vehicle purchases: buyer funds are held in the segregated structure and released to the seller only upon a defined release event — vehicle release confirmation (Module F) or expiry of the assurance-claim window for HMP Assured lots — with dispute freeze capability (Trust & Safety hold).
- **FR-E-013 (M):** Payouts to sellers via Faster Payments with Confirmation of Payee verification; payout accounts require penny-drop or Open Banking ownership verification; first-payout and changed-account payouts pass an enhanced fraud review gate.

E.3 Trade Credit

- **FR-E-020 (M):** Trade credit application workflow: KYB (Companies House + UBO identification), commercial credit bureau integration (Experian/Creditsafe class) with automated decisioning bands (auto-approve / refer / decline), credit-limit assignment, and periodic (quarterly) automated re-assessment with limit adjustment events.
- **FR-E-021 (M):** Credit exposure engine: real-time available-credit computation consumed by bid validation (FR-D-011); overdue invoices automatically restrict bidding entitlement (grace rules configurable); dunning workflow with staged communications and late-payment fee assessment per published terms.

E.4 Fee Computation & Multi-Party Distribution

- **FR-E-030 (M): Fee engine:** a declarative, versioned rules system computing buyer and seller fees per transaction from dimensions including: sale channel, hammer/sale price (banded sliding scales), account tier/subscription, lot type (fleet/dealer/private entry), assurance products, payment method, and promotional overrides. Every computed fee line stores the rule-set version and full input snapshot (deterministic replay requirement).
- **FR-E-031 (M):** Representative fee structures to be supported at launch: banded buyer's premium (e.g., percentage with min/max caps per price band), seller entry fee + commission, listing/ad-product charges, assurance fee, late-payment fees, transport charges (pass-through + margin), and subscription billing (recurring, metered overage).
- **FR-E-032 (M): VAT engine:** correct treatment per line — standard-rated fees, **VAT margin scheme** vs. VAT-qualifying vehicle sales, zero-rated exports with evidence capture — producing MTD-compatible VAT reporting artifacts. Vehicle VAT status is a mandatory catalog attribute validated at lot entry.

- **FR-E-033 (M):** Invoice generation: legally compliant sales invoices and self-billing agreements where applicable, generated within 60 s of the triggering event, delivered via portal + email (PDF/A + structured e-invoice payload), with credit-note workflow for adjustments (never destructive edits).
- **FR-E-034 (M):** Multi-party distribution: on settlement of a sale, the engine shall atomically post the split — seller proceeds, platform commission, VAT control, assurance reserve fund, transport partner payable, referral/introducer commission where present — as one balanced journal transaction.

E.5 Financing Integration (Consumer)

- **FR-E-040 (M):** Finance calculators (PCP/HP/PCH) embedded on listings with lender-panel rate cards, FCA CONC-compliant representative examples, APR disclosure, and total-amount-payable displays. All calculator display logic shall be centrally versioned for compliance sign-off (no per-page drift).
- **FR-E-041 (M):** Finance application flow: soft-search eligibility check (no credit-score footprint until consumer proceeds), consented handoff to lender panel via broker API, decision webhook ingestion (approved/refer/declined + terms), and commission attribution back to the transaction record. The Platform operates as a credit broker, not lender — status disclosures mandatory on all finance surfaces.
- **FR-E-042 (S):** Part-exchange settlement handling: outstanding-finance figure retrieval, settlement quote validity windows, and equity/negative-equity computation surfaced in the deal builder.

E.6 Reconciliation & Controls

- **FR-E-050 (M):** Automated daily three-way reconciliation (internal ledger ↔ PSP settlement reports ↔ bank statements via Open Banking feeds) with exception queue, aging, and materiality-tiered alerting; unreconciled items > 48 h escalate to Finance Operations leadership.
- **FR-E-051 (M):** Four-eyes approval on: manual journal entries, refunds above threshold, fee-schedule publication, payout-account overrides. All financial admin actions are covered by RBAC audit (REQ RBAC-004).

Module F — Post-Sale Operations & Logistics

- **FR-F-001 (M): Transport booking:** integrated quote-and-book across a carrier panel (API-integrated transport partners + internal fleet), with quotes computed from distance, vehicle class, drive/non-runner status, and open/enclosed requirements; buyer self-serve booking at checkout or post-sale; consolidated multi-vehicle moves for trade buyers.
- **FR-F-002 (M):** Logistics job lifecycle: Quoted → Booked → Assigned → Collected → InTransit → Delivered → PODConfirmed, with carrier mobile workflow (collection condition photos, e-signature, GPS breadcrumbs), exception codes (failed collection, damage-in-transit), and customer tracking page with ETA.
- **FR-F-003 (M):** Damage-in-transit claims: collection vs. delivery photo comparison pack auto-assembled, claim intake within a configurable window (default 48 h post-delivery), and liability routing (carrier vs. seller vs. assurance fund).
- **FR-F-010 (M): Vehicle release control:** a lot shall be physically/logically releasable only upon a signed **Release Note** generated when: payment cleared (or credit-covered), identity of collector verified (QR-based release code + photo ID capture at gate), and no active holds (dispute, provenance re-check failure). Gate-app integration for auction-site security.

- **FR-F-011 (M): Title/V5C transfer tracking:** workflow state per vehicle — logbook received/verified at ingestion, keeper-change notification (DVLA online keeper transfer integration where available; tracked postal fallback), and completion confirmation — with SLA dashboards and automated chase communications; a listing/lot shall disclose V5C status (present/applied-for) as a mandatory catalog attribute.
 - **FR-F-012 (M):** Document vault per transaction: invoices, release note, condition report version-of-record, MOT/service documents, POD, and transfer confirmations — retained per statutory schedule and downloadable by entitled parties only.
 - **FR-F-020 (M): Assurance/dispute workflow** (HMP Assured, FR-A-024): structured claim intake within the covered window (default 48 business hours post-collection), evidence capture, triage SLAs, resolution actions (price adjustment, rejection/return, repair contribution) with ledger integration for adjustments (credit notes, fund drawdown), and outcome analytics feeding inspector quality scoring.
 - **FR-F-021 (S):** Post-purchase consumer journey: handover scheduling, digital handover checklist, warranty product upsell integration, and review solicitation (verified-purchase gated).
-

4. UI/UX Strategy & User Journey Flows

4.1 Design System Mandates

- **UX-001 (M):** A single, token-based design system (“HMP Design Language”) shall govern all surfaces — responsive web, iOS, Android, and internal consoles — implemented as: design tokens (color, type scale, spacing, elevation, motion) exported from Figma as the source of truth, a themable component library (web: React; mobile: shared component kit), and automated visual-regression testing on every component release.
- **UX-002 (M):** Dual-audience theming: a **Consumer theme** (editorial, imagery-led, generous whitespace) and a **Trade theme** (dense, data-forward, dark-mode-first for auction halls and evening bidding) built from the same token architecture.
- **UX-003 (M):** Accessibility: WCAG 2.2 AA across all consumer surfaces, including full keyboard operability of the live bidding interface, screen-reader announcements for bid events (ARIA live regions, rate-limited to remain intelligible), and reduced-motion alternatives for countdown/urgency animations.
- **UX-004 (M):** The live bidding surface shall follow safety-critical UI patterns: bid buttons display the exact commit amount (“Bid £14,250” — never a bare “Bid” label), enforce a 300 ms debounce with optimistic-but-labeled pending state, position destructive/commit actions away from navigation gestures, and require explicit confirmation above a configurable large-bid threshold.
- **UX-005 (S):** Localization-ready copy architecture (ICU message format, no concatenated strings) even though launch is UK-only.

4.2 Platform Surface Requirements

- **UX-010 (M): Responsive web:** mobile-first, server-rendered for all SEO-bearing pages; Core Web Vitals budgets: LCP ≤ 2.0 s (P75, 4G mid-tier device), INP ≤ 200 ms, CLS ≤ 0.1 ; search-results interactions (facet toggle, pagination) shall not trigger full page reloads.
- **UX-011 (M): Native mobile apps (iOS/Android):** full marketplace and trade-bidding capability; push notification categories (outbid, lot-starting-soon, saved-search match, offer received, payout sent) individually controllable; biometric re-auth gate before bid placement and payments; offline-tolerant catalog browsing with clearly staled-data indicators (never stale *bid* state — live surfaces require connectivity and say so).
- **UX-012 (M): Inspector app:** offline-first field operation (full inspection capture without connectivity, background sync with conflict rejection), guided shot list (FR-A-014), and hardware integration (Bluetooth OBD dongles, tread-depth gauges where API-capable).
- **UX-013 (M): Auctioneer rostrum console:** dedicated large-format layout, latency-instrumented controls (FR-D-034), hardware failover (secondary device hot-standby with one-tap lane takeover).

4.3 User Journey — Listing a Car (Private Seller)

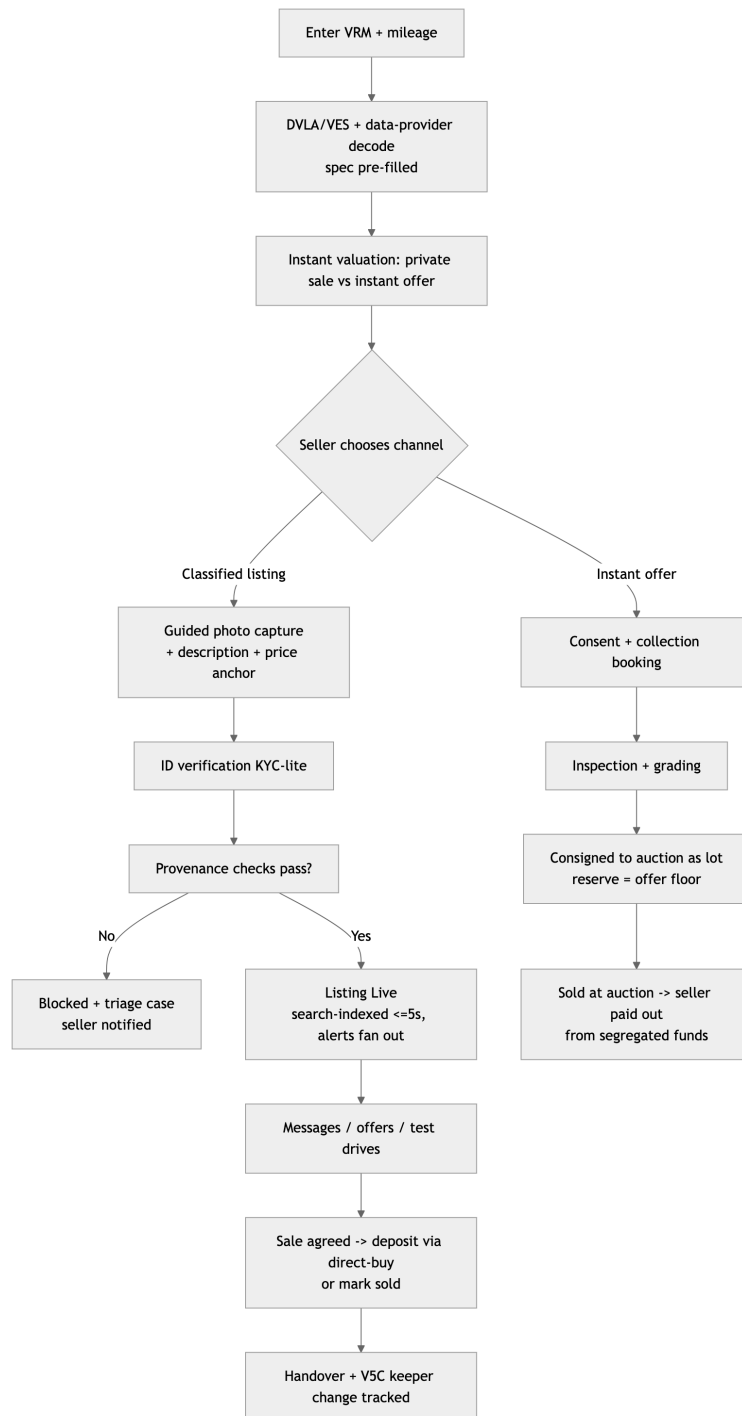


Figure 1 — User Journey: Listing a Car (Private Seller)

4.4 User Journey — Bidding on a Car (Trade Buyer)

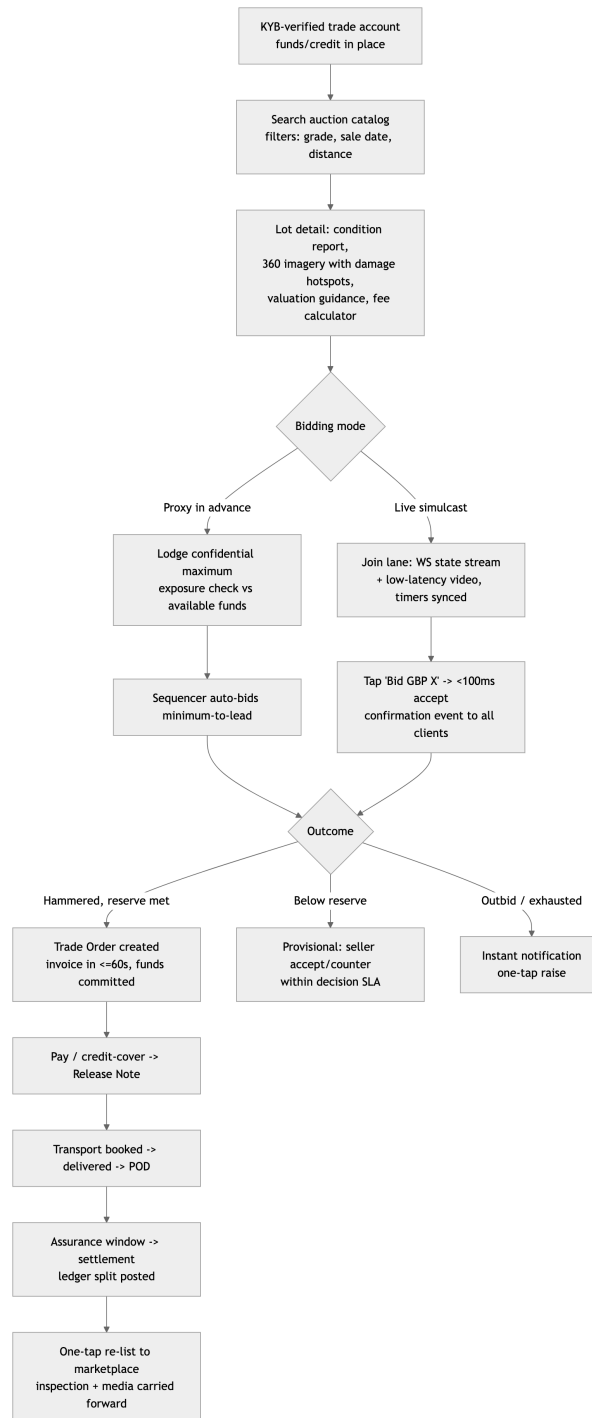


Figure 2 — User Journey: Bidding on a Car (Trade Buyer)

4.5 Journey-Level Experience Requirements

- **UX-020 (M):** Time-to-first-listing (private seller, VRM entry → live pending checks) ≤ 8 minutes at P50, instrumented as a product KPI.

- **UX-021 (M):** The lot detail page shall present a **total-cost calculator** (hammer scenario + buyer fees + VAT + transport estimate) before any bid can be placed — no fee surprises (supports FCA Consumer Duty outcomes for retail-facing surfaces and trade transparency generally).
- **UX-022 (M):** Cross-channel continuity: a saved vehicle, search, or armed proxy shall be visible and actionable identically on web and mobile within 5 s of creation (state sync via user-event stream).

5. Non-Functional Requirements (NFRs)

5.1 Performance

ID	Requirement	Target	Priority
NFR-P-001	Bid processing (ingress → sequenced decision → confirmation emitted), intra-region	< 100 ms P99 at peak	M
NFR-P-002	Bid state fan-out to all subscribed bidder clients (server emit → client receive), same region	< 150 ms P95	M
NFR-P-003	Search API server-side latency (standard query envelope)	≤ 300 ms P95	M
NFR-P-004	Listing detail page server response (cache-warm)	≤ 200 ms P95	M
NFR-P-005	Search index freshness — retail mutations / auction lot state	≤ 5 s / ≤ 1 s	M
NFR-P-006	Simulcast video glass-to-glass	≤ 3 s (bid data never gated on video)	M
NFR-P-007	Timer synchronization skew across clients	≤ ±150 ms	M
NFR-P-010	Core Web Vitals (P75, mid-tier 4G): LCP / INP / CLS	≤ 2.0 s / ≤ 200 ms / ≤ 0.1	M

5.2 Scalability & Availability

- **NFR-A-001 (M):** Peak auction concurrency envelope (design load): **50 simultaneous lanes**, 25,000 concurrently connected trade clients, 150,000 concurrent spectators, sustained 500 accepted bids/second platform-wide with burst to 2,000/s — with headroom demonstrated at 3× design load in pre-production load tests before each capacity-relevant release.
- **NFR-A-002 (M):** Availability SLOs: Auction Core **99.95%** during scheduled sale windows (error budget policy: feature freeze on exhaustion); marketplace read path 99.9%; financial ledger writes 99.95% monthly.
- **NFR-A-003 (M):** The Auction Core shall run cell-isolated per lane-group: a failure in one cell shall not affect other lanes; marketplace traffic spikes (e.g., TV-ad surges) shall be architecturally incapable of degrading bidding (separate compute pools, brokers, and datastores).
- **NFR-A-004 (M):** Sequencer failover: on bid-sequencer instance loss, a standby shall resume from the replicated bid log with ≤ 2 s of lot pause and zero accepted-bid loss (RPO = 0 for the bid log); the auctioneer console shall surface the pause explicitly.
- **NFR-A-005 (M):** DR posture: multi-AZ active-active in-region; cross-region warm standby with RTO ≤ 30 min, RPO ≤ 1 min for transactional stores (ledger RPO = 0 via synchronous journal replication policy).
- **NFR-A-006 (M):** Graceful degradation ladder (documented and chaos-tested quarterly): recommendations → personalization → media renditions → search sorting sophistication shed first; bid path and ledger shed last (never).
- **NFR-A-007 (S):** Elastic scale automation keyed to the sale calendar: pre-warm capacity 60 min before scheduled sales rather than reactive-only autoscaling.

5.3 Security & Compliance

- **NFR-S-001 (M): UK GDPR/DPA 2018:** lawful-basis mapping per processing purpose, consent management (granular, PECR-compliant for marketing/tracking), DSAR automation (export/erasure with financial-record statutory-retention carve-outs), DPIAs for valuation profiling, fraud scoring, and CV damage detection; data-minimized pseudonymization in analytics stores.
- **NFR-S-002 (M): PCI-DSS v4.0** at SAQ-A scope (FR-E-011): PSP-hosted capture only; quarterly ASV scans; segmentation evidence maintained.
- **NFR-S-003 (M): FCA compliance:** credit-broking permissions, CONC-compliant financial promotions (centrally version-controlled disclosure components), Consumer Duty outcome monitoring (price/value dashboards for finance & assurance products), and complaints handling (DISP) workflow with FOS referral rights disclosure.
- **NFR-S-004 (M): AML/KYC-KYB:** risk-based due diligence for trade accounts and high-value private transactions; PEP/sanctions screening (initial + ongoing delta screening); suspicious-activity monitoring rules (structuring of deposits, rapid buy-sell loops, third-party payment attempts) with SAR case-management support; cash handling excluded by policy (no cash acceptance $\geq$ regulatory thresholds — digital payment rails only).
- **NFR-S-005 (M):** Authentication: OIDC/OAuth 2.1 via central IdP; MFA mandatory for trade, dealer-admin, and all internal roles; WebAuthn/passkeys supported for all users; step-up authentication for bids above threshold, payout changes, and credit drawdown.

- **NFR-S-006 (M):** Encryption: TLS 1.3 in transit (internal mTLS via service mesh); AES-256 at rest with KMS-managed keys, per-datastore key separation, and field-level envelope encryption for reserves, proxy maxima, bank details, and identity documents.
- **NFR-S-007 (M): Fraud detection:** real-time scoring on account creation, listing publication, bid placement, and payout mutation — signals include device fingerprinting, velocity rules, shill-bidding detection (bidder-seller association graphs, bid-pattern anomaly models), payment-fraud signals from PSP, and image-reuse detection across listings (known-scam imagery hashes). Actions: friction (step-up), hold, or block, with human review queue and appeal path.
- **NFR-S-008 (M):** Secure engineering baseline: OWASP ASVS L2 (L3 for payment & bidding services), dependency and container scanning in CI with SLA-ed remediation, secrets in a managed vault (no secrets in code/config), annual CREST-accredited penetration test plus pre-launch test of the Auction Core, and a public responsible-disclosure policy.
- **NFR-S-009 (M):** Anti-abuse for market integrity: rate limiting per user/device/IP with trade-aware allowances, bot management on search/scraping surfaces (licensed data-feed products offered as the sanctioned alternative), and auction-specific protections (bid-flooding throttles that never delay the first valid bid).
- **NFR-S-014 (S):** Data-residency architecture: UK/EU region pinning for personal data, with tenancy metadata designed so future non-UK expansion does not require re-architecture.

5.4 Observability, Auditability & Operability

- **NFR-O-001 (M):** Distributed tracing (OpenTelemetry) across every request path with trace-ID propagation into audit events; the bid path shall be traced at 100% sampling during sale windows.
- **NFR-O-002 (M):** SLO dashboards and burn-rate alerting per service; a dedicated **Sale-Day Operations** console aggregating lane health, bid latency percentiles, WS connection churn, and PSP status.
- **NFR-O-003 (M):** Immutable audit trail (append-only, WORM storage class) for: RBAC decisions, financial postings, reserve/lot mutations, and admin actions — retention ≥ 7 years.
- **NFR-O-004 (M):** Feature-flag and progressive-delivery infrastructure (canary + automated rollback on SLO regression) for all services; auction-core changes additionally gated behind sale-calendar-aware deployment freezes.

6. Proposed Technical Architecture & Infrastructure Stack

6.1 Architecture Style

Event-driven microservices with Domain-Driven Design boundaries and CQRS/event-sourcing where the domain demands it (Auction Core, Ledger). Guiding decisions:

1. **Two planes, one platform:** The **Marketplace Plane** (read-heavy, cache-friendly, SEO-bound) and the **Auction Plane** (write-ordered, latency-critical, stateful) are deployed in separate compute cells sharing only the event backbone and canonical Vehicle Master data — enforcing NFR-A-003 at the topology level, not by convention.

2. **Event backbone as system of integration:** All domain facts (ListingPublished , BidAccepted , LotHammered , InvoiceIssued , VehicleReleased) flow through a durable log (Kafka), giving search indexing, analytics, notifications, and the valuation model a consistent, replayable feed. Services own their data; no shared databases across bounded contexts.
3. **Event sourcing where truth is a sequence:** The bid log and the financial journal are append-only event-sourced aggregates (zero-loss, replayable, natively auditable — directly satisfying FR-D-041, FR-E-001, NFR-A-004). Marketplace CRUD domains use conventional state persistence with change-data-capture into the backbone.
4. **Edge-to-core real-time path:** WebSocket gateway tier (connection management, auth, backpressure) → bid command router → **single-writer lot sequencer** (in-memory state machine per lot) → fan-out tier (bidder-priority vs. spectator). This is the < 100 ms path: a dedicated **Laravel Octane (Swoole)** worker set holding lot state in shared-memory Swoole Tables, persisting the bid log to a **Redis Streams append-only log** (AOF-fsync'd, replica-backed) with asynchronous archival to PostgreSQL — long-lived coroutine workers, zero per-request framework bootstrap on the hot path.

High-Level Topology

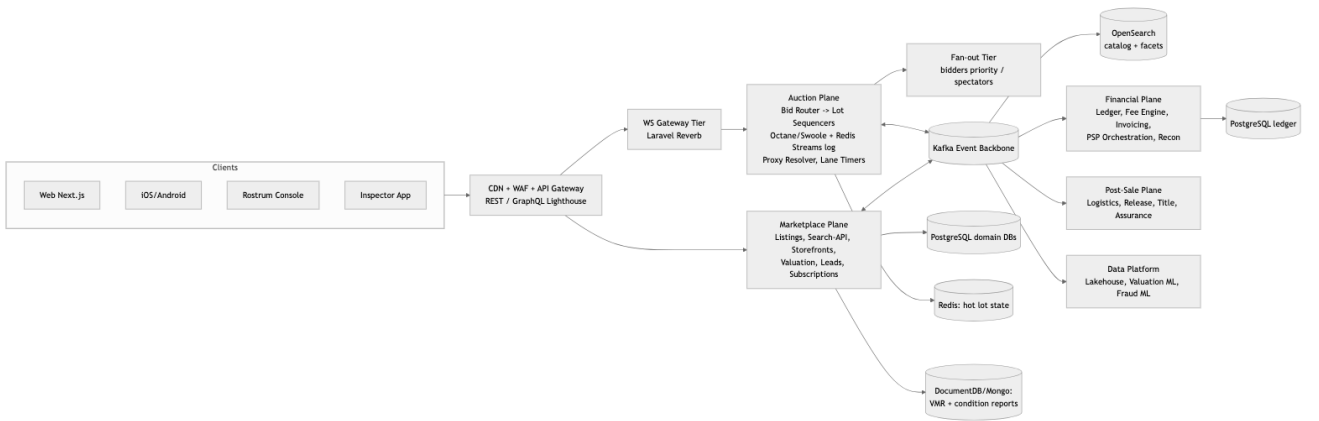


Figure 3 — High-Level System Topology

6.2 Frontend & Mobile

Layer	Selection	Rationale
Web framework	TypeScript + React 19 + Next.js (App Router) — SSR/ISR for SEO pages, React Server Components for search/detail, islands of client interactivity for live surfaces	Meets FR-C-004 SEO mandates and NFR-P-010 budgets with one stack
Live-bidding web client	Dedicated WS client library built on Laravel Echo over Reverb (shared TS package) with sequence-gap detection, snapshot resync, idempotent bid submission, and clock-offset module (FR-D-030/031/032)	Protocol correctness centralized, reused across surfaces; first-party pairing with the Reverb gateway
Mobile	React Native (New Architecture) + TypeScript , sharing the domain/WS	

Layer	Selection	Rationale
	packages with web; native modules for camera-guided capture, biometrics, and OBD (inspector)	Team leverage + shared bidding protocol; native performance where it matters
Design system	Token pipeline (Style Dictionary) → React web components + RN components; Storybook + Chromatic visual regression	UX-001 enforcement
Internal consoles	Same React stack, Trade theme, deployed behind zero-trust access	Consistency, lower TCO

6.3 Backend & Real-Time Services

Backend framework baseline: PHP 8.4 + Laravel 12 across all backend services. Latency-critical and long-running services run under **Laravel Octane (Swoole)**; conventional CRUD services run standard FPM. One framework, one dependency toolchain (Composer), uniform testing/static-analysis stack — with service boundaries preserved so any single service can be re-platformed independently if scale ever demands it.

Concern	Selection	Rationale
Auction Core (sequencer, bid router, proxy resolver, timers)	Dedicated Laravel Octane (Swoole) service: single-writer-per-lot coroutine workers, lot state in shared-memory Swoole Tables , append-only bid log in Redis Streams (AOF-fsync'd, replica-backed) with async archival to PostgreSQL; proxy resolver executes inside the sequencer to preserve total ordering (FR-D-021)	Long-lived workers with no per-request bootstrap meet the P99 < 100 ms budget; Redis Streams replication gives NFR-A-004 failover with RPO 0
WS gateway & fan-out	Laravel Reverb , horizontally scaled behind the load balancer with a Redis pub/sub backplane ; presence channels; bidder-priority vs. spectator channel tiers (FR-D-036); clients connect via Laravel Echo	First-party Laravel WebSocket server; 175k-connection envelope via horizontal scale-out
Domain services (listings, storefronts, leads, subscriptions, logistics, identity)	Laravel 12 modular services — one deployable per bounded context; Eloquent with an explicit repository/query layer; Laravel Horizon -supervised Redis queues for async work; Laravel Scheduler for recurring jobs (feed ingestion, alert fan-out, reconciliation runs)	Uniform, rapid delivery across transactional business domains; mature first-party tooling
Financial plane (ledger, fee engine, invoicing)	Laravel services over PostgreSQL with serializable-isolation journal writes inside database transactions; brick/money for arbitrary-precision monetary arithmetic; fee rules as versioned	Correctness-first; deterministic replay; no floating-point money

Concern	Selection	Rationale
	declarative artifacts (FR-E-030); domain events published via a transactional outbox	
Event backbone integration	Kafka producers/consumers via php-rdkafka ; every service publishes through its transactional outbox; consumers run as Horizon-supervised long-running daemon workers	Exactly-once-effect semantics from PHP onto the Kafka backbone
ML services (valuation, fraud, CV damage)	Python (FastAPI) model-serving behind the gateway; feature store fed from Kafka; batch training on the lakehouse	Standard MLOps toolchain (unchanged by the Laravel baseline)
API layer	REST-first — Laravel API Resources with OpenAPI specs (Scribe-generated) ; GraphQL via Lighthouse where client-side composition pays for itself; signed webhooks (native Laravel) for partner/dealer integrations; AsyncAPI-documented WS protocol	Right protocol per consumer, all from one framework
AuthZ/AuthN	OIDC IdP (Keycloak or Auth0) as the central identity provider; Laravel Sanctum for first-party SPA/mobile token issuance behind the IdP; RBAC/ABAC enforced via Laravel Policies/Gates + spatie/laravel-permission , with policy definitions synchronized from the central PDP (REQ RBAC-002) so decisions stay centrally auditable	Central identity, framework-native enforcement at every service
Video simulcast	Managed WebRTC/LL-HLS provider (Millicast/IVS class) — video is decoupled from the bid data channel (FR-D-035)	Buy, don't build, non-differentiating media transport

6.4 Database Layer — Polyglot Persistence

Store	Technology	Owned by / Used for
Transactional financial ledger	PostgreSQL (append-only journal tables, serializable isolation, synchronous replica; partitioned by period)	Ledger, invoices, fee computations — the money truth (FR-E-001)
Domain relational stores	PostgreSQL (one logical DB per service)	Accounts/orgs, subscriptions, orders, logistics jobs, credit
Bid log	Redis Streams append-only log within the Auction Core (AOF-fsync'd, replica-backed); archived to object storage	FR-D-010, FR-D-041

Store	Technology	Owned by / Used for
	post-sale; projections to PostgreSQL for reporting	
Hot real-time state	Redis (clustered)	Live lot state snapshots, WS session registry, rate limiting, exposure counters
Catalog & condition documents	MongoDB/DocumentDB via the official <code>mongodb/laravel-mongodb</code> package (Eloquent-compatible)	Vehicle Master Records, versioned condition reports, media manifests — deep, schema-flexible documents
Search	OpenSearch/Elasticsearch , integrated via Laravel Scout with a custom OpenSearch engine driver for faceted queries	Faceted retail + auction catalog indexes (FR-B-001), alerting percolation for saved searches
Event backbone	Apache Kafka (with schema registry, Avro/Protobuf contracts)	System of integration; CDC via Debezium from domain DBs
Analytics lakehouse	Object storage + Iceberg/Delta tables, Spark/Trino, dbt	Valuation & fraud training data, BI, regulatory reporting
Object storage	S3-class with lifecycle policies + WORM buckets	Media masters/renditions, documents vault, audit archive (NFR-O-003)

6.5 Infrastructure & Delivery

- **Cloud & runtime:** AWS (eu-west-2 primary, eu-west-1 warm standby), **Kubernetes (EKS)** with separate node pools/cells for Auction vs. Marketplace planes; Karpenter autoscaling with sale-calendar pre-warming (NFR-A-007). PHP containers built on hardened FPM/Octane base images with OPcache preloading and JIT tuned per service class.
- **Edge:** CloudFront CDN + WAF + bot management; image optimization at edge; API Gateway with per-persona rate-limit classes.
- **laC & delivery:** Terraform (modules per environment), GitOps (Argo CD), trunk-based development, progressive delivery (canary + automated SLO-based rollback via Argo Rollouts), ephemeral preview environments per PR for web surfaces.
- **Service mesh & zero trust:** mTLS everywhere (Istio/Linkerd), workload identity (SPIFFE-class), egress control; secrets via Vault/KMS.
- **Observability stack:** OpenTelemetry → managed backend (Datadog/Grafana LGTM), RUM on web/mobile, synthetic bid-path probes running continuously against a heartbeat lane.
- **Quality gates:** Pest/PHPUnit suites with mutation testing on financial code (Infection), **PHPStan/Larastan (max level)** static analysis and **Laravel Pint** formatting enforced in CI, contract tests (Pact) on every service boundary, deterministic replay tests for the fee engine and proxy state machine (golden event-log fixtures), chaos experiments for the degradation ladder (NFR-A-006), and 3× design-load performance tests as a release gate for the Auction Core (NFR-A-001).

6.6 Build vs. Buy Summary

Capability	Decision	Candidate
Bid sequencing & proxy engine	Build (core differentiator)	—
Ledger & fee engine	Build (differentiating correctness) over PSP primitives	—
Payments/PSP & client-money safeguarding	Buy	Adyen / Stripe + EMI partner
Vehicle data, provenance, valuations (licensed baseline)	Buy	CAP HPI / Autovista / Experian AutoCheck class
Identity verification (KYC/KYB)	Buy	Onfido/Persona + Companies House API
Live video transport	Buy	IVS / Millicast class
Search engine	Adopt OSS	OpenSearch
Credit bureau	Buy	Experian / Creditsafe
Transport carrier network	Hybrid	API panel + internal fleet

7. Requirements Traceability & Acceptance

- **TRC-001 (M):** Every FR/NFR in this document shall be mirrored in the delivery tracker with bidirectional links (requirement ↔ epic ↔ test evidence); release sign-off requires 100% Must-requirement coverage with passing acceptance evidence.
- **TRC-002 (M):** NFR conformance is release-gated by automated evidence: load-test reports (NFR-A-001), latency SLO dashboards over the soak window (NFR-P-001..007), and security scan/pen-test attestations (NFR-S-008).
- **TRC-003 (M):** Compliance requirements (NFR-S-001..004) require named-owner sign-off from Compliance & Risk prior to each major release.

Assumptions & Dependencies

1. Commercial agreements with a vehicle-data provider, PSP/EMI partner, credit bureau, and IDV vendor are concluded before Module A/E build completion.
2. FCA credit-broking authorization (or appointed-representative arrangement) is obtained before consumer finance surfaces go live.
3. Physical auction sites (for simulcast) provide network redundancy meeting a defined site-connectivity SLA (dual uplinks, LTE failover).
4. DVLA/DVSA API access agreements and volume tiers are in place for VES and MOT History integration.

Open Items for Architecture Review Board

#	Item	Owner	Target
1	Octane runtime bake-off for the sequencer — Swoole vs. FrankenPHP (worker mode) — validated against the NFR-P-001 latency budget at 3× design load	Head of Platform Eng	Sprint 4
2	Lighthouse GraphQL vs. BFF-per-surface for client API composition	Principal Architect	Sprint 5
3	EMI partner shortlist for client-money safeguarding	CFO / Compliance	Sprint 6
4	Timed-online vs. auctioneer-led launch sequencing (product)	CPO	Sprint 6

End of document — HMP-SRS-001 v1.1.0. This SRS is a living baseline; changes follow the change-control process in the Document Control section and require re-approval at version increments.